

evaluate_pam50

March 27, 2026

1 Molecular Subtype Classification of TCGA-BRCA Whole-Slide Images

1.1 Problem Definition

We address the task of **PAM50 molecular subtype classification** from H&E-stained whole-slide images (WSI) in the TCGA-BRCA cohort.

The classification target consists of **4 molecular subtypes**:

Class	Label	Description
0	Luminal A	Hormone receptor-positive, low proliferation
1	Luminal B	Hormone receptor-positive, higher proliferation
2	Basal-like	Triple-negative, aggressive
3	HER2-enriched	HER2 overexpression

Slides labelled as *Normal-like* are excluded from training and evaluation, consistent with standard practice.

Dataset statistics (after exclusion): 986 diagnostic slides from TCGA-BRCA, distributed as:

- Luminal A: 508 slides (51.5%)
- Luminal B: 222 slides (22.5%)
- Basal-like: 173 slides (17.5%)
- HER2-enriched: 83 slides (8.4%)

1.2 Pipeline

The classification pipeline follows a two-stage **Multiple Instance Learning (MIL)** approach:

1.2.1 Stage 1: Patch Feature Extraction – UNI2-h

Each WSI is tessellated into non-overlapping 224x224 patches at 20x magnification. Patch-level features are extracted using **UNI2-h** (Mahmood Lab), a ViT-Giant/14 foundation model pre-trained on over 100M histopathology images. Each patch is represented as a **1536-dimensional** embedding vector. The resulting bag of embeddings per slide is stored as an HDF5 file.

1.2.2 Stage 2: Slide-Level Classification – CLAM-MB

Slide-level predictions are made using **CLAM-MB** (Clustering-constrained Attention Multiple Instance Learning, Multi-Branch variant). CLAM-MB uses gated attention to learn which patches are most informative for each class and aggregates them into a slide-level representation. The model is trained with:

- **Bag-level loss:** Cross-entropy with label smoothing (0.1)
- **Instance-level loss:** SmoothTop1SVM (auxiliary clustering constraint)
- **Weighted random sampling** to address class imbalance
- **Early stopping** on validation AUC (patience = 5 epochs)

1.3 Experimental Setup

1.3.1 Hyperparameter Optimization

Hyperparameters were tuned via **Bayesian optimization** (W&B Sweeps), maximizing **validation AUC** on fold 0. The search space included:

Parameter	Search Space
Learning rate	Log-uniform [1e-4, 1e-3]
Weight decay	Log-uniform [5e-4, 1e-2]
Dropout	{0.25, 0.5, 0.6, 0.75}
Model variant	{CLAM-SB, CLAM-MB}
Attention size	{small, big}
Bag weight	Uniform [0.7, 0.95]
Top-k patches (B)	{4, 8, 16}
Instance loss	SVM (fixed)
Early stopping patience	{3, 5, 7}

1.3.2 Best Configuration (selected by validation AUC)

Parameter	Value
Model	CLAM-MB, big
Learning rate	1.008e-4
Weight decay	2.446e-6
Dropout	0.5
Bag weight	0.553
Top-k patches (B)	4
Instance loss	SVM
Optimizer	Adam
Max epochs	50
Early stopping patience	5

1.3.3 Evaluation Protocol

The best configuration was evaluated using **stratified 10-fold cross-validation** with patient-level stratification. Each slide appears in exactly one test fold, so aggregating predictions across all folds yields held-out predictions for the entire dataset without data leakage.

1.4 Results

1.4.1 Per-Fold Validation and Test Metrics

Table 1 shows per-fold AUC (macro one-vs-rest) and overall accuracy for both validation and test sets.

Table 1: Per-Fold Results

	test_auc	val_auc	test_acc	val_acc
Fold				
0	0.9197	0.9250	0.6699	0.7885
1	0.8669	0.8577	0.7234	0.6354
2	0.8487	0.9044	0.6598	0.7670
3	0.9217	0.9132	0.7879	0.7500
4	0.8916	0.9082	0.7576	0.7677
5	0.8942	0.8999	0.7292	0.7053
6	0.9015	0.9214	0.6796	0.7576
7	0.9099	0.8944	0.7263	0.7475
8	0.9150	0.8925	0.7677	0.7789
9	0.8602	0.9030	0.6832	0.7449

test_auc	Mean +/- Std: 0.8930 +/- 0.0260			
val_auc	Mean +/- Std: 0.9019 +/- 0.0188			
test_acc	Mean +/- Std: 0.7184 +/- 0.0441			
val_acc	Mean +/- Std: 0.7443 +/- 0.0445			
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1.4.2 Aggregated Predictions

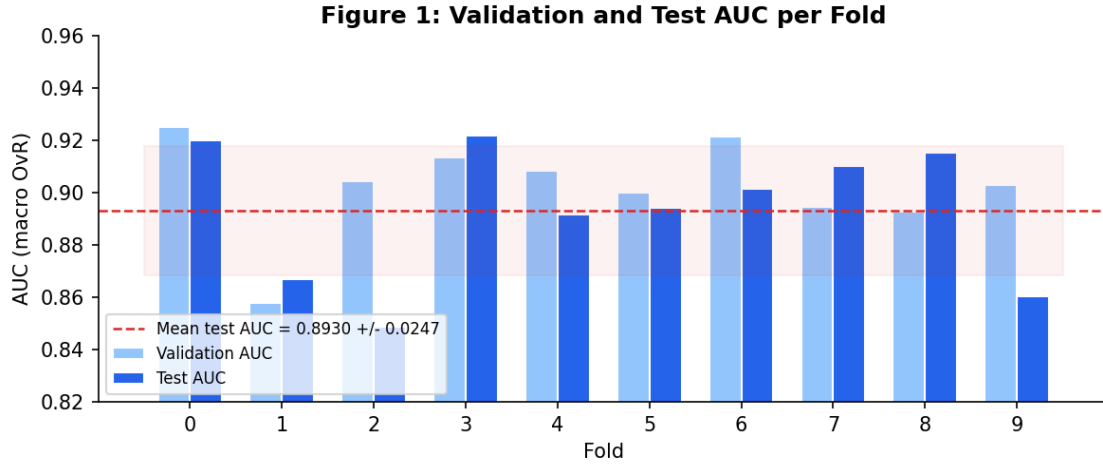
Since each slide appears in exactly one test fold, we aggregate all per-slide test predictions across the 10 folds to obtain held-out predictions for the full dataset.

Total slides evaluated: 986

Class distribution: {'LumA': 508, 'LumB': 222, 'Basal': 173, 'Her2': 83}

1.4.3 Per-Fold Test AUC

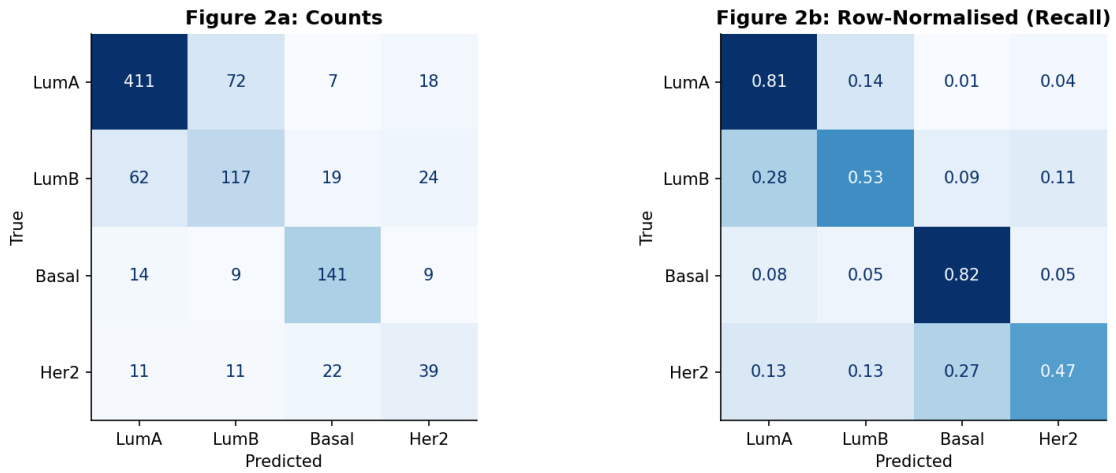
Figure 1 compares validation and test AUC across the 10 folds. The dashed red line and shaded band show the mean +/- 1 std of the test AUC.



1.4.4 Confusion Matrix (Aggregated Test Predictions)

Figure 2 shows the confusion matrix for all 986 held-out predictions, both as raw counts (left) and row-normalised recall (right).

Figure 2: Confusion Matrix -- Aggregated 10-Fold Test Predictions



1.4.5 Classification Report (Aggregated Test Predictions)

Table 2: Classification Report -- Aggregated 10-Fold Test Predictions

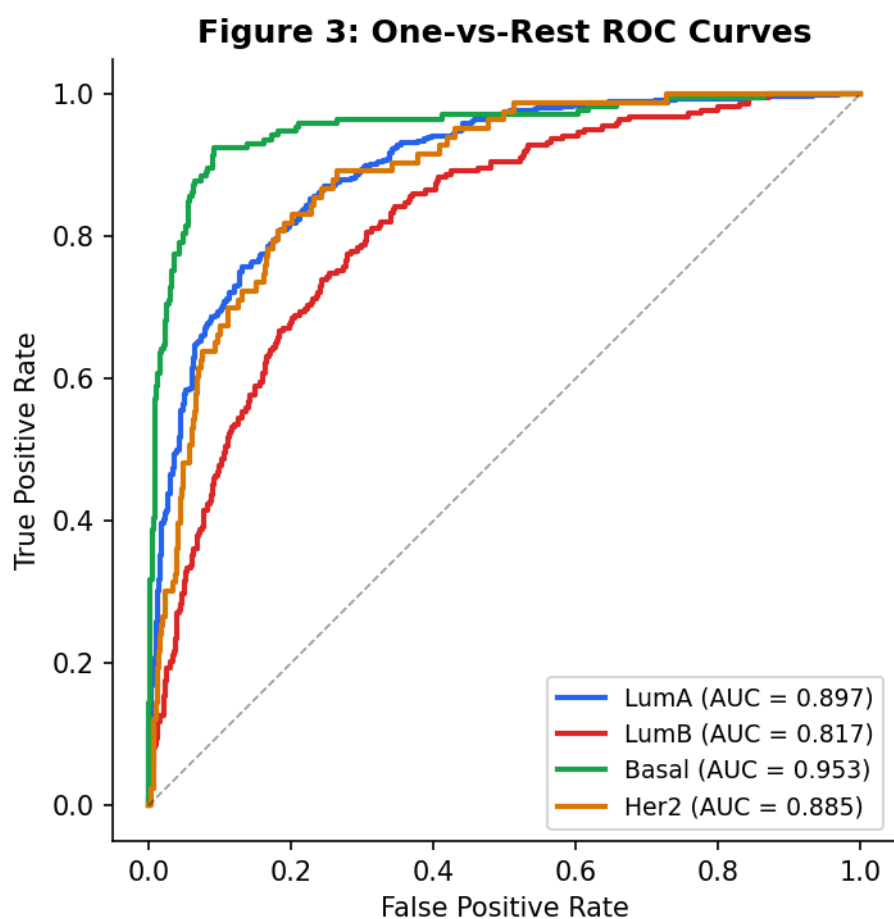
	precision	recall	f1-score	support
LumA	0.83	0.81	0.82	508
LumB	0.56	0.53	0.54	222

Basal	0.75	0.82	0.78	173
Her2	0.43	0.47	0.45	83
accuracy			0.72	986
macro avg	0.64	0.66	0.65	986
weighted avg	0.72	0.72	0.72	986

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1.4.6 ROC Curves (Aggregated Test Predictions)

Figure 3 shows one-vs-rest ROC curves for each molecular subtype computed on the aggregated held-out predictions.



1.4.7 Per-Class AUC Across Folds

Table 3 and Figure 4 break down the test AUC by class for each fold.

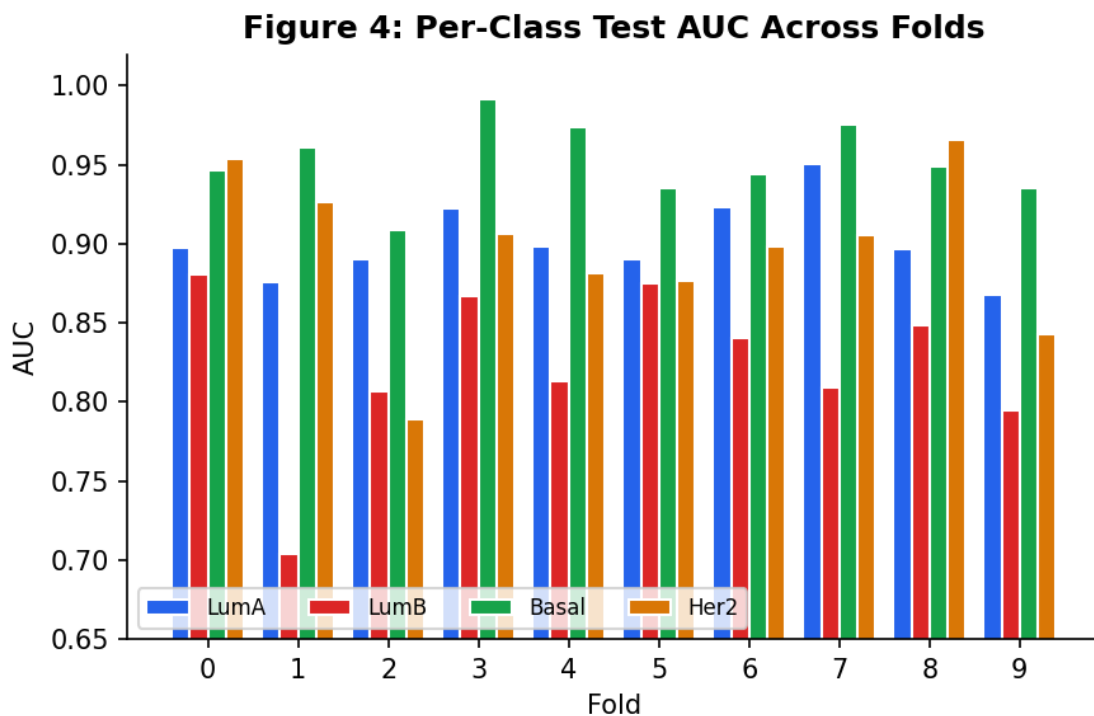
Table 3: Per-Class Test AUC by Fold

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	LumA	LumB	Basal	Her2
Fold				
0	0.8977	0.8805	0.9466	0.9539
1	0.8762	0.7041	0.9607	0.9268
2	0.8905	0.8064	0.9088	0.7893
3	0.9220	0.8666	0.9914	0.9066
4	0.8980	0.8129	0.9739	0.8815
5	0.8903	0.8750	0.9352	0.8764
6	0.9230	0.8405	0.9439	0.8987
7	0.9503	0.8088	0.9755	0.9052
8	0.8971	0.8484	0.9491	0.9657
9	0.8678	0.7946	0.9352	0.8430

Mean +/- Std:

LumA : 0.9013 +/- 0.0243
LumB : 0.8238 +/- 0.0521
Basal : 0.9520 +/- 0.0240
Her2 : 0.8947 +/- 0.0518



1.5 Summary

The table below summarises the final results of the **CLAM-MB + UNI2-h** pipeline on TCGA-BRCA molecular subtype classification using 10-fold cross-validation.

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CLAM-MB + UNI2-h | PAM50 Subtyping | 10-Fold CV
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--- Validation Set ---
Macro AUC (mean +/- std): 0.9019 +/- 0.0188
Accuracy (mean +/- std): 0.7443 +/- 0.0445

--- Test Set (aggregated held-out predictions) ---
Macro AUC (aggregated): 0.8879
Macro AUC (mean +/- std): 0.8930 +/- 0.0260
Overall Accuracy: 0.7181
Weighted F1-Score: 0.7179

--- Per-Class Test Results ---
Class      AUC      Prec  Recall      F1      n
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LumA      0.8967    0.825   0.809    0.817    508
LumB      0.8174    0.560   0.527    0.543    222
Basal     0.9529    0.746   0.815    0.779    173
Her2      0.8846    0.433   0.470    0.451     83
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